

IMSC-2025-Using derivatives in inequalities and AC-GC method

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Convex and Concave:

If $f'' \geq 0$, the function is **convex**; if $f'' \leq 0$, it is **concave**. Alternatively, if f' is non-decreasing, the function is convex; if f' is non-increasing, it is concave.

The following statements are equivalent:

1. f is convex.
2. $-f$ is concave.
3. For any x, y and $0 < t < 1$, the inequality

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$$

holds. This is a necessary and sufficient condition for f to be convex.

4. For any x, y , the inequality

$$f\left(\frac{x+y}{2}\right) \leq \frac{f(x) + f(y)}{2}$$

holds. This is a necessary and sufficient condition for f to be convex.

5. For any $x < y < z$, the inequalities

$$\frac{f(y) - f(x)}{y - x} \leq \frac{f(z) - f(x)}{z - x} \leq \frac{f(z) - f(y)}{z - y}$$

hold. This is a necessary and sufficient condition for f to be convex.

From these, the most important inequality is **Jensen's Inequality**.

Jensen's Inequality:

Let f be a convex function and n a natural number. Then, for any real numbers $x_1, x_2, \dots, x_n$ and non-negative real numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ such that $\alpha_1 + \alpha_2 + \dots + \alpha_n = 1$, the following inequality holds:

$$\alpha_1 f(x_1) + \alpha_2 f(x_2) + \dots + \alpha_n f(x_n) \geq f(\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n).$$

1. **Problem:** Let $x_1, x_2, \dots, x_n$ be positive numbers then

$$A(x_1, \dots, x_n) = \frac{x_1 + \dots + x_n}{(1 + x_1^n) \dots (1 + x_n^n)} < n$$

And if $1 < n$ then find the maximum value of $A(x_1, \dots, x_n)$.

Arithmetic; Geometric Compensation Method

Arithmetic Compensation Theorem: Let $s > 0$ and let $F(x_1, x_2, \dots, x_n)$ be a symmetrical continuous function on the compact set in $\mathbb{R}^n$

$$S = \{(x_1, x_2, \dots, x_n) \mid x_1 + x_2 + \dots + x_n = s, x_1 \geq 0, \dots, x_n \geq 0\}.$$

If

$$F(x_1, x_2, x_3, \dots, x_n) \leq \max \left\{ F\left(\frac{x_1 + x_2}{2}, \frac{x_1 + x_2}{2}, x_3, \dots, x_n\right), F(0, x_1 + x_2, x_3, \dots, x_n) \right\} \quad (1)$$

for all $(x_1, x_2, \dots, x_n) \in S$ with $x_1 > x_2 > 0$, then

$$F(x_1, x_2, x_3, \dots, x_n) \leq \max_{1 \leq k \leq n} F\left(\frac{s}{k}, \dots, \frac{s}{k}, 0, \dots, 0\right) \quad (2)$$

for all $(x_1, x_2, \dots, x_n) \in S$.

1. **Problem:** If $a, b, c, d \geq 0$ such that $a + b + c + d = 4$ then

$$(1 + 3a)(1 + 3b)(1 + 3c)(1 + 3d) \leq 125 + 131abcd$$

2. **Problem:** Let $x_1, x_2, \dots, x_n$ be non-negative real numbers such that

$$x_1 + x_2 + \dots + x_n = \frac{2}{3}$$

then

$$\sum_{1 \leq i < j \leq n} \frac{x_i x_j}{(1 - x_i)(1 - x_j)} \leq \frac{1}{4}$$

Geometric Compensation Theorem: Let $f(t)$ be a continuous function defined on $[0, \infty)$ such that for any (x, y) with $x > y > 0$, the inequality holds:

$$f(x) + f(y) \leq \max\{2f(\sqrt{xy}), a + b\},$$

where $a = f(0)$ and $b = \lim_{t \rightarrow \infty} f(t)$. Let $p > 0$, let $x_1, x_2, \dots, x_n$ be positive numbers such that:

$$x_1 x_2 \dots x_n = G^n,$$

let $k_1, k_2 \in \{1, 2, \dots, n-1\}$ and let:

$$\delta = \max_{\substack{k_1 + k_2 \leq n \\ c \geq 0}} [k_1 a + k_2 b + (n - k_1 - k_2) f(c)].$$

Then,

$$f(x_1) + f(x_2) + \dots + f(x_n) \leq \max\{\delta, n f(G)\}.$$

1. **Problem:** Let $x_1, x_2, \dots, x_n$ be positive numbers satisfying

$$\sqrt[n]{x_1 x_2 \dots x_n} = G \leq \frac{1}{n-1}$$

then

$$\frac{1}{1+x_1} + \frac{1}{1+x_2} + \dots + \frac{1}{1+x_n} \leq \frac{n}{1+G}$$